

UAKIT, Russian Federation

WMO: 304550

Lat: 55.486N

Lon: 113.619E

Elev: 1096

StdP: 88.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.6	-36.5	-41.1	0.1	-38.3	-39.2	0.1	-36.3	10.0	-17.6	8.3	-20.3	0.2	140	0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	24.9	15.4	23.2	14.9	21.5	14.2	16.9	22.0	16.0	21.0	15.1	20.0	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.4	18.6	14.1	11.5	17.8	13.0	10.7	17.2	51.9	21.9	48.8	21.0	45.9	20.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.1	7.0	DB	-41.2	28.2	2.5	1.5	-43.0	29.3	-44.5	30.2	-45.9	31.0	-47.7	32.2	(4)
(5)			WB	-41.2	19.1	2.5	1.5	-43.0	20.1	-44.5	21.0	-45.9	21.8	-47.7	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.7	-27.0	-22.6	-14.7	-4.6	4.4	12.6	15.2	12.4	4.9	-5.6	-18.3	-26.3	(6)	
	DBStd	15.86	5.36	5.41	6.08	4.98	4.27	3.76	2.81	3.20	4.26	5.76	6.82	5.74	(7)	
	HDD10.0	6099	1147	912	764	439	180	18	2	15	162	484	850	1126	(8)	
	HDD18.3	8788	1405	1146	1023	689	432	175	103	184	404	742	1100	1384	(9)	
	CDD10.0	359	0	0	0	0	6	94	162	89	7	0	0	0	(10)	
	CDD18.3	6	0	0	0	0	0	1	4	0	0	0	0	0	(11)	
	CDH23.3	138	0	0	0	0	3	45	74	17	0	0	0	0	(12)	
	CDH26.7	12	0	0	0	0	0	4	7	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.1	1.0	1.5	2.3	3.0	3.4	2.5	2.2	2.2	2.5	2.1	1.5	1.0	(14)	
(15) Precipitation	PrecAvg	385	3	3	4	16	30	73	91	94	44	12	7	4	(15)	
	PrecMax	530	9	8	8	42	93	175	194	183	117	51	25	17	(16)	
	PrecMin	176	0	0	0	3	6	22	22	26	13	0	3	1	(17)	
	PrecStd	92	2	2	2	10	21	41	41	42	23	10	4	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.5	-5.8	3.7	12.2	21.9	26.9	27.6	25.5	20.8	9.7	1.8	-8.3	(19)	
		MCWB	-13.1	-8.1	-0.7	4.5	10.5	14.9	17.3	16.0	11.5	3.9	-1.4	-9.6	(20)	
	2%	DB	-14.4	-9.1	0.0	8.2	18.2	24.5	25.3	22.8	17.0	6.7	-2.7	-12.6	(21)	
		MCWB	-15.5	-10.7	-3.4	2.3	8.6	14.4	16.2	14.9	10.1	2.2	-4.7	-13.6	(22)	
	5%	DB	-17.1	-11.9	-2.7	5.7	15.3	22.6	23.6	20.8	14.4	4.4	-6.1	-15.5	(23)	
		MCWB	-17.8	-13.1	-5.5	0.6	7.2	13.7	15.5	13.9	8.6	0.6	-7.4	-16.4	(24)	
	10%	DB	-19.4	-14.4	-5.0	3.5	12.7	20.6	22.0	18.9	12.1	2.3	-8.8	-18.1	(25)	
		MCWB	-19.9	-15.3	-7.4	-0.7	5.7	13.0	15.1	12.8	7.3	-0.8	-10.0	-18.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.0	-8.0	-0.6	5.1	11.2	17.2	18.5	17.1	12.7	4.5	-1.1	-9.7	(27)	
		MCDB	-11.6	-6.0	3.3	11.8	20.8	22.4	24.1	21.6	18.4	9.2	1.5	-8.6	(28)	
	2%	WB	-15.4	-10.7	-3.4	2.6	9.2	15.7	17.5	15.8	10.9	2.5	-4.8	-13.6	(29)	
		MCDB	-14.4	-9.2	-0.2	7.7	16.7	21.7	22.6	21.0	15.3	6.3	-2.8	-12.6	(30)	
	5%	WB	-17.8	-13.1	-5.5	1.0	7.7	14.6	16.6	14.8	9.4	0.8	-7.4	-16.4	(31)	
		MCDB	-17.1	-11.9	-2.8	5.1	14.3	20.4	21.6	19.4	13.5	3.9	-6.1	-15.6	(32)	
	10%	WB	-19.9	-15.3	-7.4	-0.5	6.2	13.7	15.8	13.9	7.8	-0.6	-9.9	-18.7	(33)	
		MCDB	-19.4	-14.5	-5.1	3.2	11.9	19.4	20.5	17.6	11.7	2.2	-8.9	-18.2	(34)	
(35) Mean Daily Temperature Range	MDBR		7.5	10.2	12.0	10.9	10.6	12.0	11.3	11.1	10.1	9.7	8.6	7.3	(35)	
	5% DB	MCDBR	9.3	12.0	13.5	13.5	16.0	16.2	15.1	14.8	14.3	11.4	10.2	9.7	(36)	
		MCWBR	8.5	10.5	10.4	8.8	8.6	7.8	7.2	7.6	7.9	7.6	8.5	8.9	(37)	
	5% WB	MCDBR	9.2	11.9	13.4	12.8	14.8	13.3	12.4	12.1	13.1	10.8	10.1	9.6	(38)	
		MCWBR	8.5	10.4	10.5	8.6	8.4	7.0	6.7	7.0	7.7	7.3	8.5	8.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.223	0.253	0.278	0.332	0.391	0.413	0.387	0.359	0.309	0.266	0.236	0.214	(40)	
	taud		2.152	2.151	2.142	2.125	2.130	2.223	2.348	2.409	2.483	2.399	2.183	2.091	(41)	
	Ebn at Noon		723	828	891	884	847	832	849	854	861	818	690	631	(42)	
	Edh at Noon		57	85	111	133	141	131	114	100	80	66	55	47	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.90	1.83	3.39	5.00	5.66	5.45	5.09	4.30	3.02	1.86	1.04	0.64	(44)	
	RadStd		0.04	0.09	0.16	0.23	0.39	0.31	0.48	0.34	0.22	0.13	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	-1.20	N/A	N/A	N/A	N/A	N/A	-3	(46)	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-3	(47)	

Nomenclature: See separate page